

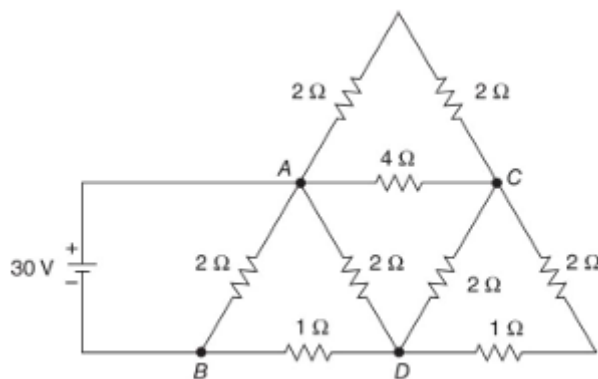
BEEE – MID – I – Important Questions

Short questions:

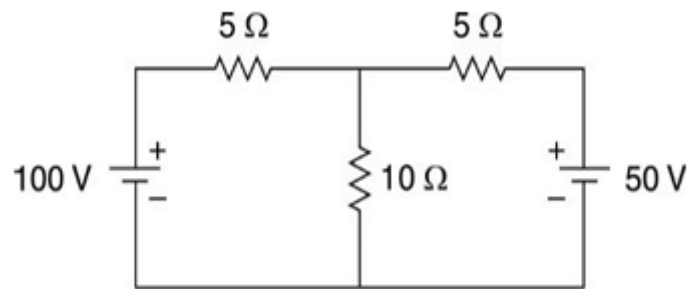
1. Define KCL.
2. Define KVL
3. State Superposition theorem.
4. State any two limitations of Ohm's Law.
5. Define Power Factor.
6. Define Impedance.
7. Define Amplitude, time period.
8. Discuss the applications of induction motors
9. What is the function of transformer.
10. State the working principle of Three phase Induction Motor.
11. Write difference between conventional and non-conventional energy sources.
12. What are the components used in nuclear power plant?
13. What is the purpose of Fuse.
14. Define "Unit".
15. What are the different types of earthing systems.

Long Questions:

1. What is ohm's law, give its limitations.
2. Determine the current delivered by the source in the circuit shown.



3. Compute the current in the 10Ω resistor as shown in Fig. using Superposition theorem.



4. Derive the R.M.S. value of a sinusoidal alternating voltage.
5. An alternating current is represented by $i = 50 \sin 314 t$. Determine (i) Average value (iii) Peak factor (ii) RMS value (iv) Form factor.
6. Explain the terms Active, Reactive, Apparent powers and Power factor.
7. Explain the Construction of DC Generator with neat sketch.
8. Explain the working principle of DC Motor.
9. Explain the principle and operation of three phase Induction Motor.
10. Describe the working principle of Alternator.
11. Explain the Construction and Operation of PMMC Instrument.
12. Explain the procedure of finding the value of an unknown resistance using Wheatstone Bridge.
13. Draw the layout and explain the operation of Hydel Power plant.
14. Describe the wind power generation with layout.
15. Explain the operation of solar power generation.
16. Explain the safety precautions to be followed to avoid electric shock.
17. What is the purpose of Earthing and discuss different types of earthing's.
18. Define Tariff and Explain about two-part electricity tariff.